

Docker Tutorial

31 July 2026 20:29

- 1.) Why Docker exists?
- 2.) What is a docker?
- 3.) Docker Architecture Explanation.
- 4.) Docker Installation.
- 5.) Containerize SpringBoot Application.
-> *Dockerfile*
->Port Mapping
->Env Config
- 6.) Docker Compose.
- 7.) Docker Volumes.
- 8.) Docker Networking.

Command	Purpose
docker --version	Check Docker version
docker images	List images
docker ps	List running containers
docker ps -a	List all containers
docker pull nginx	Download an image
docker run hello-world	Test Docker installation
docker run -d -p 8080:80 nginx	Run Nginx in background
docker stop my-nginx	Stop a container
docker start my-nginx	Start a stopped container
docker rm my-nginx	Remove a container
docker rmi nginx	Remove an image
docker run -d nginx	Detach from the container
Docker run -it ubuntu	Run ubuntu on interactive mode

On powershell ls -lart wont run it only runs ls
Nginx-> your container name which is present :)

Use the docker registry to pull the images and official website to explore the images

Powershell commands:

Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\deepa> docker -v
Docker version 29.6.2, build dfc4efb
PS C:\Users\deepa> docker images
```

i Info → U In Use

```
IMAGE ID          DISK USAGE  CONTENT SIZE  EXTRA
PS C:\Users\deepa> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
```

4f55086f7dd0: Pull complete
Digest: sha256:c3cbe1cc1aa588a64951ac6286e0df7b27fe2e6324b1001c619bb358770c0178
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
\$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
<https://hub.docker.com/>

For more examples and ideas, visit:
<https://docs.docker.com/get-started/>

```
PS C:\Users\deepa> docker compose version
Docker Compose version v5.3.1
PS C:\Users\deepa> docker run hello-world
```

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
\$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
<https://hub.docker.com/>

For more examples and ideas, visit:
<https://docs.docker.com/get-started/>

```
PS C:\Users\deepa> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
PS C:\Users\deepa> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED      STATUS      PORTS      NAMES
2b66f15eeaf4   hello-world  "/hello"  18 seconds ago  Exited (0)  17 seconds ago  goofy_pike
12bd9f868f6c   hello-world  "/hello"  30 seconds ago  Exited (0)  29 seconds ago  exciting_turing
```

```
PS C:\Users\deepa> ls -lart
Get-ChildItem : A parameter cannot be found that matches parameter name 'lart'.
At line:1 char:4
+ ls -lart
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls -last
Get-ChildItem : A parameter cannot be found that matches parameter name 'last'.
At line:1 char:4
+ ls -last
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls -l-ast
Get-ChildItem : A parameter cannot be found that matches parameter name 'last'.
At line:1 char:4
+ ls -l-ast
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls -lart
Get-ChildItem : A parameter cannot be found that matches parameter name 'lart'.
At line:1 char:4
+ ls -lart
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls -latt
Get-ChildItem : A parameter cannot be found that matches parameter name 'latt'.
At line:1 char:4
+ ls -latt
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls -lart
Get-ChildItem : A parameter cannot be found that matches parameter name 'lart'.
At line:1 char:4
+ ls -lart
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId :
NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand
```

```
PS C:\Users\deepa> ls
```

Directory: C:\Users\deepa

Mode	LastWriteTime		Length	Name
----	-----	-----	----	----
d----	10-12-2023	14:39		.anaconda
d----	21-03-2025	00:22		.android
d----	02-02-2023	15:43		.arduinoIDE
d----	24-08-2024	10:37		.astropy
d----	06-02-2024	08:08		.aws
d----	16-07-2025	10:10		.azure
d----	08-04-2025	14:32		.cache
d----	08-05-2025	18:47		.conda
d----	01-08-2026	11:56		.config
d----	10-12-2023	14:39		.continuum
d----	17-02-2026	22:08		.copilot
d----	21-03-2025	22:45		.cursor
d----	01-08-2026	11:57		.docker
d----	16-07-2025	10:18		.dotnet
d----	18-07-2024	16:52		.eclipse
d----	23-07-2026	13:57		.gk
d----	07-12-2024	08:14		.gradle
d----	03-07-2023	21:29		.idlerc
d----	07-08-2025	07:41		.ipynb_checkpoints
d----	09-05-2023	13:32		.ipython
d----	25-06-2025	21:09		.jupyter
d----	08-04-2025	13:46		.keras
d----	21-02-2024	12:45		.kube
d----	01-08-2024	16:48		.lemminx
d----	26-07-2024	21:16		.m2
d----	17-07-2024	17:00		.matplotlib
d----	21-02-2024	12:39		.minikube
d----	09-01-2023	18:32		.nbi
d----	16-07-2025	10:18		.nuget
d----	27-08-2022	09:28		.Origin
d----	29-07-2026	12:17		.p2
d----	27-08-2022	09:28		.QtWebEngineProcess
d----	07-01-2024	18:58		.quokka
d----	31-01-2025	14:14		.spyder-py3
d----	28-07-2026	13:05		.ssh
d----	14-11-2024	11:27		.sts4
d----	23-07-2025	09:55		.templateengine
d----	17-07-2024	21:42		.VirtualBox
d----	13-06-2023	10:46		.vscode
d----	06-01-2024	20:34		.wallaby
d----	29-09-2024	11:28		apache-tomcat-7.0.109
d----	16-07-2025	10:13		CalcApp
d-r--	26-12-2022	22:03		Contacts
d----	07-05-2026	12:54		DaVinci_Resolve_20.3.2_Windows
d----	15-02-2024	14:34		dem
d----	15-12-2023	07:37		demobit
d-r--	29-04-2024	10:23		Desktop
d----	29-07-2026	12:16		Documents
d-r--	01-08-2026	11:29		Downloads
d----	14-12-2023	09:58		drop
d----	09-06-2025	09:37		eclipse
d----	08-04-2026	19:25		eclipse-workspace

d-r---	26-12-2022	22:03	Favorites
d----	27-09-2024	12:18	jboss-eap-7.2
d----	10-05-2023	19:24	Jedi
d----	12-01-2024	08:10	junit5
d-r---	26-12-2022	22:03	Links
d----	23-10-2022	09:50	Microsoft
d-r---	07-05-2026	12:29	Music
d----	24-11-2025	12:15	MyApp
d----	02-03-2026	10:58	node_modules
d----	04-05-2024	14:14	npm-cache
dar--l	29-10-2023	17:33	OneDrive
d----	21-03-2025	11:25	Postman Agent
d----	24-07-2025	18:52	PycharmProjects
d-r---	24-07-2024	15:01	Searches
d----	08-04-2026	19:45	SpringJobPortal
d----	08-02-2024	14:19	test
d-r---	23-03-2025	06:12	Videos
d----	14-03-2024	14:15	VirtualBox VMs
d----	16-12-2024	12:19	workspace
-a----	01-08-2026	11:52	95 .accessibility.properties
-a----	29-04-2024	10:50	8323 .bash_history
-a----	26-04-2024	00:02	213 .gitconfig
-a----	01-11-2024	11:36	0 .junit5session
-a----	25-07-2025	08:00	20 .lessht
-a----	17-03-2025	16:09	12 .node_repl_history
-a----	24-07-2025	18:41	8129 banglore_home_prices_model.pickle
-a----	24-07-2025	14:57	938020 Bengaluru_House_Data.csv
-a----	24-07-2025	15:27	744720 bhp.csv
-a----	16-06-2025	21:23	8915 Book1.xlsx
-a----	16-07-2025	10:12	1117 calculator.cs
-a----	16-12-2024	12:48	1615 calculators.html
-a----	03-07-2025	12:14	378 carprices.csv
-a----	27-07-2025	20:17	338485 claim_or_not.ipynb
-a----	24-07-2025	18:41	3994 columns.json
-a----	23-06-2025	16:35	384 customers-100.csv
-a----	08-07-2025	21:13	50399 decision_tree.ipynb
-a----	05-08-2025	07:30	63040138 DEvideos.csv
-a----	08-07-2025	12:19	103459 digit_dataser.ipynb
-a----	01-08-2026	11:30	643194800 Docker Desktop Installer.exe
-a----	16-10-2023	12:02	0 Github
-a----	05-08-2025	07:51	161515 highest_reach.ipynb
-a----	03-07-2025	12:19	352 homeprices.csv
-a----	17-07-2025	19:23	363 income.csv
-a----	08-07-2025	11:32	155 insurance_data.csv
-a----	16-07-2025	19:19	6247 k_folds.ipynb
-a----	22-07-2025	08:04	82416 k_mean_clustering.ipynb
-a----	14-05-2026	19:29	1501 LICENCE
-a----	08-07-2025	11:52	56261 logistic_regression.ipynb
-a----	23-06-2025	17:02	116 main.csv
-a----	24-06-2025	16:59	135 main_term.csv
-a----	22-07-2025	08:25	59646 naive_bais.ipynb
-a----	22-07-2025	19:46	30158 naivy_bais_part2.ipynb
-a----	02-03-2026	10:58	18927 package-lock.json
-a----	02-03-2026	10:58	131 package.json
-a----	24-07-2025	15:01	38670 pca.ipynb
-a----	14-05-2026	19:28	1573 README.txt
-a----	24-07-2025	18:43	321440 Real_estate.ipynb

-a----	11-07-2025	11:49	1748 RF.ipynb
-a----	08-07-2025	20:56	638 salaries.csv
-a----	22-07-2025	19:17	480130 spam.csv
-a----	11-07-2025	11:46	86355 SVM.ipynb
-a----	24-11-2025	12:16	765 swap.cs
-a----	24-11-2025	12:17	780 swaping.cs
-a----	24-11-2025	12:16	765 swaps.exe
-a----	27-07-2025	20:09	3082519 tiktok_dataset.csv
-a----	22-07-2025	08:07	60302 titanic.csv
-a----	24-06-2025	16:34	171741 Untitled.ipynb
-a----	23-06-2025	17:04	135678 Untitled1.ipynb
-a----	23-06-2025	17:07	3364 Untitled2.ipynb
-a----	24-06-2025	16:58	29023 Untitled3.ipynb
-a----	25-06-2025	21:46	165460 Untitled4.ipynb
-a----	03-07-2025	13:13	83192 Untitled5.ipynb
-a----	08-07-2025	20:29	617 Untitled6.ipynb
-a----	21-07-2025	18:53	72 Untitled7.ipynb
-a----	26-07-2025	08:35	2826 Untitled8.ipynb
-a----	07-08-2025	07:41	72 Untitled9.ipynb
-a----	16-07-2025	10:08	1117 using System;.cs
-a----	03-07-2023	18:03	0 version
-a----	14-05-2026	19:28	104 website.url

PS C:\Users\deepa> docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

PS C:\Users\deepa> docker pull ninrod/springboot:test

Error response from daemon: unsupported manifest media type and no default available:
application/vnd.docker.distribution.manifest.v1+prettyjws

PS C:\Users\deepa> docker pull springboot/maven-cache-resource

Using default tag: latest

latest: Pulling from springboot/maven-cache-resource

cfc728c1c558: Pull complete

5b12b87f0a0e: Pull complete

b7d6497a92f9: Pull complete

56c8963b1109: Pull complete

0ee5b9859ab4: Pull complete

b6df9668cc31: Pull complete

1fb9f56e8e3e: Pull complete

a50490627e81: Pull complete

8c2e1734baf7: Pull complete

7df8dea0dba8: Pull complete

e11b6e435b26: Pull complete

9af4c238c901: Pull complete

24e11564717d: Pull complete

953176780efd: Pull complete

Digest: sha256:6b62ece54058e6ca3361ad7b7ea08f090acca085c36f761ef29958d286d8a496

Status: Downloaded newer image for springboot/maven-cache-resource:latest

docker.io/springboot/maven-cache-resource:latest

What's next:

View a summary of image vulnerabilities and recommendations → docker scout quickview
springboot/maven-cache-resource

PS C:\Users\deepa> docker -help

Flag shorthand -h has been deprecated, use --help

unknown shorthand flag: 'e' in -elp

Usage: docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information

```
PS C:\Users\deepa> docker -help
```

Flag shorthand -h has been deprecated, use --help

unknown shorthand flag: 'e' in -elp

Usage: docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information

```
PS C:\Users\deepa> docker --help
```

Usage: docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:

run	Create and run a new container from an image
exec	Execute a command in a running container
ps	List containers
build	Build an image from a Dockerfile
bake	Build from a file
pull	Download an image from a registry
push	Upload an image to a registry
images	List images
login	Authenticate to a registry
logout	Log out from a registry
search	Search Docker Hub for images
version	Show the Docker version information
info	Display system-wide information

Management Commands:

agent*	Docker AI Agent Runner
ai*	Docker AI Agent - Ask Gordon
builder	Manage builds
buildx*	Docker Buildx
compose*	Docker Compose
container	Manage containers
context	Manage contexts
debug*	Get a shell into any image or container
desktop*	Docker Desktop commands
dhi*	CLI for managing Docker Hardened Images
extension*	Manages Docker extensions
image	Manage images
init*	Creates Docker-related starter files for your project
manifest	Manage Docker image manifests and manifest lists
mcp*	Docker MCP Plugin
model*	Docker Model Runner
network	Manage networks
offload*	Docker Offload
pass*	Docker Pass Secrets Manager Plugin (beta)
plugin	Manage plugins
scout*	Docker Scout
system	Manage Docker
volume	Manage volumes

Swarm Commands:

swarm	Manage Swarm
-------	--------------

Commands:

attach Attach local standard input, output, and error streams to a running container
commit Create a new image from a container's changes
cp Copy files/folders between a container and the local filesystem
create Create a new container
diff Inspect changes to files or directories on a container's filesystem
events Get real time events from the server
export Export a container's filesystem as a tar archive
history Show the history of an image
import Import the contents from a tarball to create a filesystem image
inspect Return low-level information on Docker objects
kill Kill one or more running containers
load Load an image from a tar archive or STDIN
logs Fetch the logs of a container
pause Pause all processes within one or more containers
port List port mappings or a specific mapping for the container
rename Rename a container
restart Restart one or more containers
rm Remove one or more containers
rmi Remove one or more images
save Save one or more images to a tar archive (streamed to STDOUT by default)
start Start one or more stopped containers
stats Display a live stream of container(s) resource usage statistics
stop Stop one or more running containers
tag Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
top Display the running processes of a container
unpause Unpause all processes within one or more containers
update Update configuration of one or more containers
wait Block until one or more containers stop, then print their exit codes

Global Options:

--config string Location of client config files (default "C:\\Users\\deepa\\.docker")
-c, --context string Name of the context to use to connect to the daemon (overrides DOCKER_HOST env var and default context set with "docker context use")
-D, --debug Enable debug mode
-H, --host string Daemon socket to connect to
-l, --log-level string Set the logging level ("debug", "info", "warn", "error", "fatal") (default "info")
--tls Use TLS; implied by --tlsverify
--tlscacert string Trust certs signed only by this CA (default "C:\\Users\\deepa\\.docker\\ca.pem")
--tlscert string Path to TLS certificate file (default "C:\\Users\\deepa\\.docker\\cert.pem")
--tlskey string Path to TLS key file (default "C:\\Users\\deepa\\.docker\\key.pem")
--tlsverify Use TLS and verify the remote
-v, --version Print version information and quit

Run 'docker COMMAND --help' for more information on a command.

For more help on how to use Docker, head to <https://docs.docker.com/go/guides/>
PS C:\Users\deepa> docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
-------	----	------------	--------------	-------

i Info → U In Use


```
hello-world:latest          e2ac70e7319a    10.1kB      0B   U
springboot/maven-cache-resource:latest 93604565439c    146MB      0B
PS C:\Users\deepa> docker containers
docker: unknown command: docker containers
```

Run 'docker --help' for more information
PS C:\Users\deepa> docker container
Usage: docker container COMMAND

Manage containers

Commands:

- attach Attach local standard input, output, and error streams to a running container
- commit Create a new image from a container's changes
- cp Copy files/folders between a container and the local filesystem
- create Create a new container
- diff Inspect changes to files or directories on a container's filesystem
- exec Execute a command in a running container
- export Export a container's filesystem as a tar archive
- inspect Display detailed information on one or more containers
- kill Kill one or more running containers
- logs Fetch the logs of a container
- ls List containers
- pause Pause all processes within one or more containers
- port List port mappings or a specific mapping for the container
- prune Remove all stopped containers
- rename Rename a container
- restart Restart one or more containers
- rm Remove one or more containers
- run Create and run a new container from an image
- start Start one or more stopped containers
- stats Display a live stream of container(s) resource usage statistics
- stop Stop one or more running containers
- top Display the running processes of a container
- unpause Unpause all processes within one or more containers
- update Update configuration of one or more containers
- wait Block until one or more containers stop, then print their exit codes

Run 'docker container COMMAND --help' for more information on a command.

```
PS C:\Users\deepa> docker run nginx:stable-alpine3.24-perl
Unable to find image 'nginx:stable-alpine3.24-perl' locally
stable-alpine3.24-perl: Pulling from library/nginx
55afa1ecc21d: Pull complete
33c8cdf1e4bc: Pull complete
86ffcbfd30a5: Pull complete
ac2d8fcab7e1: Pull complete
5728898b7b71: Pull complete
8f9abe9abe21: Pull complete
1bf9957874ad: Pull complete
f42b2e963e9e: Pull complete
f2a2a80c11ed: Pull complete
Digest: sha256:154841870d6ad1a16efd200063ab6671798eed2cd5f1448db5c8bdb1751111d8
Status: Downloaded newer image for nginx:stable-alpine3.24-perl
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
```

10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/01 06:44:12 [notice] 1#1: using the "epoll" event method
2026/08/01 06:44:12 [notice] 1#1: nginx/1.30.4
2026/08/01 06:44:12 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/08/01 06:44:12 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/08/01 06:44:12 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/08/01 06:44:12 [notice] 1#1: start worker processes
2026/08/01 06:44:12 [notice] 1#1: start worker process 30
2026/08/01 06:44:12 [notice] 1#1: start worker process 31
2026/08/01 06:44:12 [notice] 1#1: start worker process 32
2026/08/01 06:44:12 [notice] 1#1: start worker process 33
2026/08/01 06:44:12 [notice] 1#1: start worker process 34
2026/08/01 06:44:12 [notice] 1#1: start worker process 35
2026/08/01 06:44:12 [notice] 1#1: start worker process 36
2026/08/01 06:44:12 [notice] 1#1: start worker process 37
2026/08/01 06:44:12 [notice] 1#1: start worker process 38
2026/08/01 06:44:12 [notice] 1#1: start worker process 39
2026/08/01 06:44:12 [notice] 1#1: start worker process 40
2026/08/01 06:44:12 [notice] 1#1: start worker process 41
2026/08/01 06:44:12 [notice] 1#1: start worker process 42
2026/08/01 06:44:12 [notice] 1#1: start worker process 43
2026/08/01 06:44:12 [notice] 1#1: start worker process 44
2026/08/01 06:44:12 [notice] 1#1: start worker process 45
2026/08/01 06:46:46 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/08/01 06:46:46 [notice] 30#30: exiting
2026/08/01 06:46:46 [notice] 31#31: exiting
2026/08/01 06:46:46 [notice] 32#32: exiting
2026/08/01 06:46:46 [notice] 42#42: exiting
2026/08/01 06:46:46 [notice] 30#30: exit
2026/08/01 06:46:46 [notice] 31#31: exit
2026/08/01 06:46:46 [notice] 42#42: exit
2026/08/01 06:46:46 [notice] 32#32: exit
2026/08/01 06:46:46 [notice] 33#33: exiting
2026/08/01 06:46:46 [notice] 36#36: exiting
2026/08/01 06:46:46 [notice] 33#33: exit
2026/08/01 06:46:46 [notice] 36#36: exit
2026/08/01 06:46:46 [notice] 35#35: exiting
2026/08/01 06:46:46 [notice] 34#34: exiting
2026/08/01 06:46:46 [notice] 37#37: exiting
2026/08/01 06:46:46 [notice] 35#35: exit
2026/08/01 06:46:46 [notice] 37#37: exit
2026/08/01 06:46:46 [notice] 39#39: exiting
2026/08/01 06:46:46 [notice] 38#38: exiting
2026/08/01 06:46:46 [notice] 39#39: exit
2026/08/01 06:46:46 [notice] 40#40: exiting
2026/08/01 06:46:46 [notice] 40#40: exit
2026/08/01 06:46:46 [notice] 41#41: exiting
2026/08/01 06:46:46 [notice] 43#43: exiting
2026/08/01 06:46:46 [notice] 44#44: exiting
2026/08/01 06:46:46 [notice] 41#41: exit
2026/08/01 06:46:46 [notice] 43#43: exit
2026/08/01 06:46:46 [notice] 44#44: exit
2026/08/01 06:46:46 [notice] 45#45: exiting

```

2026/08/01 06:46:46 [notice] 45#45: exit
2026/08/01 06:46:46 [notice] 38#38: exit
2026/08/01 06:46:46 [notice] 34#34: exit
2026/08/01 06:46:46 [notice] 1#1: signal 14 (SIGALRM) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 36
2026/08/01 06:46:46 [notice] 1#1: worker process 36 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 43 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: signal 29 (SIGIO) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 41
2026/08/01 06:46:46 [notice] 1#1: worker process 34 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 38 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 39 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 41 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: signal 29 (SIGIO) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 39
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/08/01 06:46:46 [notice] 1#1: worker process 30 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: signal 29 (SIGIO) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 33
2026/08/01 06:46:46 [notice] 1#1: worker process 31 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 32 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 33 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 45 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: signal 29 (SIGIO) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 32
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 37
2026/08/01 06:46:46 [notice] 1#1: worker process 37 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: signal 29 (SIGIO) received
2026/08/01 06:46:46 [notice] 1#1: signal 17 (SIGCHLD) received from 40
2026/08/01 06:46:46 [notice] 1#1: worker process 35 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 40 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 42 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: worker process 44 exited with code 0
2026/08/01 06:46:46 [notice] 1#1: exit
PS C:\Users\deepa> docker run -d nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
062e450697fa: Pull complete
82454cdbf456: Pull complete
3c7ab7949321: Pull complete
cacfcdd01f30: Pull complete
b6698f04e005: Pull complete
2bedaf25031a: Pull complete
d26f27cc8c41: Pull complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
e1439c60987c78fd04c8912524e3eda172871977ed7b26ea62453414a232bce3
PS C:\Users\deepa> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
e1439c60987c   nginx    "/docker-entrypoint...." 55 seconds ago Up 54 seconds 80/tcp      gifted_shtern
PS C:\Users\deepa> docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS      NAMES
435bcf004051   nginx:stable-alpine3.24-perl "/docker-entrypoint...." 26 seconds ago Up 26 seconds 80/tcp      boring_hermann
e1439c60987c   nginx                "/docker-entrypoint...." 2 minutes ago  Up 2 minutes  80/tcp

```

```

gifted_shtern
PS C:\Users\deepa> docker stop 435bcf004051
435bcf004051
PS C:\Users\deepa> docker ps
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
e1439c60987c  nginx  "/docker-entrypoint...."  3 minutes ago  Up 3 minutes  80/tcp
gifted_shtern
PS C:\Users\deepa> docker stop e1439c60987c
e1439c60987c
PS C:\Users\deepa> docker ps
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
PS C:\Users\deepa> docker ps -a
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
435bcf004051  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  2 minutes ago  Exited (0) 53 seconds ago  boring_hermann
e1439c60987c  nginx  "/docker-entrypoint...."  4 minutes ago  Exited (0) 17 seconds ago  gifted_shtern
e5d68b44184a  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  7 minutes ago  Exited (0) 4 minutes ago  determined_buck
2b66f15eeaf4  hello-world  "/hello"  22 minutes ago  Exited (0) 22 minutes ago  goofy_pike
12bd9f868f6c  hello-world  "/hello"  22 minutes ago  Exited (0) 22 minutes ago  exciting_turing
PS C:\Users\deepa> docker start 12bd9f868f6c
12bd9f868f6c
PS C:\Users\deepa> docker ps -a
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
435bcf004051  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  3 minutes ago  Exited (0) 2 minutes ago  boring_hermann
e1439c60987c  nginx  "/docker-entrypoint...."  5 minutes ago  Exited (0) About a minute ago  gifted_shtern
e5d68b44184a  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  8 minutes ago  Exited (0) 6 minutes ago  determined_buck
2b66f15eeaf4  hello-world  "/hello"  23 minutes ago  Exited (0) 23 minutes ago  goofy_pike
12bd9f868f6c  hello-world  "/hello"  24 minutes ago  Exited (0) 9 seconds ago  exciting_turing
PS C:\Users\deepa> docker stop 12bd9f868f6c
12bd9f868f6c
PS C:\Users\deepa> docker ps -a
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
435bcf004051  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  4 minutes ago  Exited (0) 2 minutes ago  boring_hermann
e1439c60987c  nginx  "/docker-entrypoint...."  6 minutes ago  Exited (0) 2 minutes ago  gifted_shtern
e5d68b44184a  nginx:stable-alpine3.24-perl  "/docker-entrypoint...."  9 minutes ago  Exited (0) 6 minutes ago  determined_buck
2b66f15eeaf4  hello-world  "/hello"  24 minutes ago  Exited (0) 24 minutes ago  goofy_pike
12bd9f868f6c  hello-world  "/hello"  24 minutes ago  Exited (0) 36 seconds ago  exciting_turing
PS C:\Users\deepa> docker image
Usage: docker image COMMAND

```

Manage images

Commands:

build Build an image from a Dockerfile
history Show the history of an image
import Import the contents from a tarball to create a filesystem image
inspect Display detailed information on one or more images
load Load an image from a tar archive or STDIN
ls List images
prune Remove unused images
pull Download an image from a registry
push Upload an image to a registry
rm Remove one or more images
save Save one or more images to a tar archive (streamed to STDOUT by default)
tag Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE

Run 'docker image COMMAND --help' for more information on a command.

```
PS C:\Users\deepa> docker pull ubuntu:stonking-20260705
stonking-20260705: Pulling from library/ubuntu
494b6931e231: Pull complete
4d5d62685fc4: Pull complete
Digest: sha256:694b773ee7e0d0b55ca74c095ac3309055589e7cbfaf3100f0b226c38c6936fa
Status: Downloaded newer image for ubuntu:stonking-20260705
docker.io/library/ubuntu:stonking-20260705
```

What's next:

View a summary of image vulnerabilities and recommendations → docker scout quickview
ubuntu:stonking-20260705

```
PS C:\Users\deepa> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
435bcf004051	nginx:stable-alpine3.24-perl	"/docker-entrypoint...."	5 minutes ago	Exited (0)	4 minutes ago
e1439c60987c	nginx	"/docker-entrypoint...."	7 minutes ago	Exited (0)	3 minutes ago
e5d68b44184a	nginx:stable-alpine3.24-perl	"/docker-entrypoint...."	10 minutes ago	Exited (0)	8 minutes ago
2b66f15eeaf4	hello-world	"/hello"	25 minutes ago	Exited (0)	25 minutes ago
12bd9f868f6c	hello-world	"/hello"	26 minutes ago	Exited (0)	2 minutes ago

```
PS C:\Users\deepa> docker image
Usage: docker image COMMAND
```

Manage images

Commands:

build Build an image from a Dockerfile
history Show the history of an image
import Import the contents from a tarball to create a filesystem image
inspect Display detailed information on one or more images
load Load an image from a tar archive or STDIN
ls List images
prune Remove unused images
pull Download an image from a registry
push Upload an image to a registry
rm Remove one or more images

save Save one or more images to a tar archive (streamed to STDOUT by default)
tag Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE

Run 'docker image COMMAND --help' for more information on a command.

PS C:\Users\deepa> docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	e2ac70e7319a	10.1kB	0B	U
nginx:latest	4e5db4761e0f	161MB	0B	U
nginx:stable-alpine3.24-perl	64788b97f1a5	103MB	0B	U
springboot/maven-cache-resource:latest	93604565439c	146MB	0B	
ubuntu:stonking-20260705	ddf17c91c25b	101MB	0B	

PS C:\Users\deepa> docker run -it ubuntu

Unable to find image 'ubuntu:latest' locally

latest: Pulling from library/ubuntu

ed819469700f: Pull complete

a3679419df18: Pull complete

Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb

Status: Downloaded newer image for ubuntu:latest

root@c8135002fb72:/# ls

bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var

root@c8135002fb72:/# touch filename.txt

root@c8135002fb72:/# ls

bin boot dev etc filename.txt home lib lib64 media mnt opt proc root run sbin srv sys tmp
usr var

root@c8135002fb72:/# exit

PS C:\Users\deepa> docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

PS C:\Users\deepa>

You can check the status of the containers, the docker desktop shows you the exact status of the containers created and and their state of running, can create , stop and delelte the conatainers using the docker commands.

What is a dockerfile?

A Dockerfile is text file that contains step-by-step instructions to create a Docker image.